Kaustubh Hemant Sonawane

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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

AI Engineer with 3 years of industry experience taking LLM and **agentic** systems from zero to production in regulated financial services, starting as the only engineer alongside my manager. Built **RAG** services and **tool-calling** agents end to end in **Python** (**FastAPI**, **PostgreSQL** with **pgvector**) with a **React** frontend, **Azure OpenAI** and **LlamaIndex**, and owned the whole loop: choosing between self-hosted **Llama-3** and **Azure OpenAI** on cost and fit, **prompt engineering** (system prompts versioned in Git and tuned from real user feedback), AWS deployment and release. **Computer vision** exposure through an image-captioning study (CNN, ResNet, ViT) and a vision model in a shipped app. Completed an MSc in Data Science (awaiting result), with a dissertation project with AstraZeneca; placed 2nd at the Encode hackathon. De facto team lead for up to 4 interns; twice recognised with Best Team Project.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

Production LLM & Agentic Systems

- Specified, designed and built the company's first **GenAI** applications end to end as the only engineer alongside my manager (**ReactJS** frontend, **Python FastAPI** microservices, **LlamaIndex** with **Azure OpenAI** GPT-3.5 & Llama-3) for Compliance, Client Service and Investment, with **Azure OpenAI embeddings** and a separate **pgvector** (**PostgreSQL**) database per department, so each team's context came only from its own documents.
- Built **tool-calling** agents that call **Python** functions as tools to run **SQL**-backed fund comparisons and generate PDF reports on request; guardrailed behaviour with **system prompts** versioned in **Git**, released to real users, changed the applications based on what they found, and had Compliance review each release.
- Weighed cost against quality when choosing models: **self-hosted Llama-3** on a **GPU EC2** instance for the largest department and the stronger **Azure OpenAI** outputs for decision makers, with the ability to switch models as needs changed.

Ownership & Delivery

- Worked directly with business stakeholders in Client Service, Compliance and Investment, moving from request to working tool to release; acted as de facto team lead for up to 4 interns in my final year; recognised with Best Team Project twice (2024 rating-email automation, 2025 department RAG chatbot).
- Built a **Python** extraction pipeline, scheduled with **EventBridge**, **Lambda** and **AWS SageMaker**, that uses **Azure OpenAI** to parse over **300** daily emails from credit rating agencies into structured data (replacing a first **AWS Textract** version that broke when an agency began sending images), fuzzy-matches company names against the ones analysts track and sends a daily summary, saving **60** analyst hours a month.

Production Infrastructure & Security

- Turned prototypes into deployable tools: packaged the **FastAPI** microservices (uvicorn) as **Docker** containers on **AWS** (**EC2**, **ECR**) and automated deployment of the separate frontend and backend repositories (dev and prod) through a one-click **Jenkins** and **GitLab CI/CD** pipeline.

- Protected the internal applications with **RBAC** using a **Python** authentication function against the company **LDAP** server and **AWS Security Groups** limiting access to company private IPs; deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to passing the security team's **VAPT**, and coordinated with them on which ports to whitelist.
- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), integrated with legacy analyst data.

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built. Data preprocessing pipeline, scoring algorithm and the RAG-based AI assistant that narrates results and answers methodology questions, in a tool built as FastAPI microservices with a React and TypeScript frontend that ranks cancer cell lines by target inclusion and exclusion genes and returns the evidence behind each pick; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **University of Bristol MSc. Applied Data Science Group Project:** VisionCaptioner (Team project. Multi-phase computer vision study of CNN, ResNet and ViT image encoders paired with symbolic, retrieval and LSTM (with attention) decoders to generate natural-language descriptions of synthetic scenes; evaluated across three datasets and five seeds with SBERT cosine, BLEU and ROUGE-L and benchmarked against pre-trained vision-language models.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **Wingman:** AI dating-coach web app with 108 LLM personas across 12 archetypes, each persona's system prompt assembled from data plus per-archetype few-shot examples so tuning needs no code change, and a vision model scoring uploaded photos (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM). LLM output treated as untrusted: validated with Pydantic, repaired and cached by input hash; rate limits shaped the design (12 calls instead of 108, results streamed to the UI). GitHub: https://github.com/Arishtanemi4/wingman.git

ACHIEVEMENTS

- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Azure OpenAI, LlamaIndex, RAG, agentic workflows, prompt engineering, tool calling, computer vision (CNN, ResNet, ViT), REST APIs, Git, Docker, Jenkins, GitLab (CI/CD), AWS (EC2 including a GPU instance, S3, ECR, Lambda, EventBridge, SageMaker), Nginx, Linux, LDAP, Claude Code (AI-assisted development), codeplain (spec-driven development), Power BI, VS Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic, uvicorn), SQL, R (dplyr, tidyverse, ggplot2), JavaScript/TypeScript (NodeJS express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector semantic search, SQL functions and procedures), SQLite